

Real-Time Production Monitoring with **IoT**

How a leading automotive Manufacturer achieved 10-15% productivity boost and 90% reduction in manual effort through connected factory automation

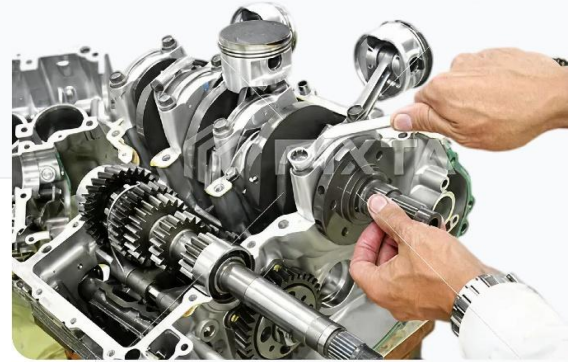
Client Overview

A leading automotive die-casting manufacturer producing high-precision aluminium components at scale. With multiple machines running continuous operations, the leadership team required live production insights to reduce downtime, increase throughput, and meet customer delivery timelines consistently.

 **Industry:** Manufacturing – Aluminium Die Casting

 **Focus:** Automotive Components

 **Scale:** Multi-machine continuous operations



Key Challenges

Traditional manufacturing processes created significant operational blind spots that prevented optimal production efficiency and timely decision-making.



Manual Data Capture

Error-prone and delayed reporting process



Inconsistent Target Tracking

Deviations spotted only during review meetings



No Live Dashboards

Production monitored post-shift; issues found too late



No Breakdown Warnings

Machines failed before detection, causing unexpected downtime



Operator Accountability Gaps

No real-time performance checks or monitoring



GWC's IoT-Driven Solution

We deployed a Connected Factory Framework enabling automated live insights across the entire production floor.



IoT + Cloud Integration

- ✓ IoT Core to ingest telemetry from machine counters
- ✓ Low-latency device communication for continuous streaming



Real-Time Data Platform

- ✓ GCP Cloud hosting pipelines and messaging services
- ✓ Streaming into Real-time DB for dashboard updates within seconds



Full-Stack Production Monitoring App

- ✓ React + Node.js application for supervisors
- ✓ Live display of machine status (running, idle, breakdown)
- ✓ Color-coded alerting for deviations



Target vs Actual Shift Insights

- ✓ Automatic tracking of hourly & shift KPIs
- ✓ Visual alerts for low-performing machines



Production Analytics Layer

- ✓ Python + SQL data processing jobs
- ✓ Operator productivity and part-rejection visibility



100% Automated Workflow

- ✓ Zero manual effort required
- ✓ Instant data capture and processing

Real Business Impact

Quantifiable improvements across key operational metrics, delivering measurable value to production efficiency and bottom-line results.



Reporting Latency

Shift-end → Real-time

From delayed post-shift reports to instant live visibility



Downtime Transparency

Instant Logging

Logged instantly with cause tagging and tracking



Productivity

10-15% ↑

Via faster intervention and proactive management



Manual Effort

90% Reduction

Automated data capture eliminates manual logging



On-Time Delivery

Improved

Enhanced through live tracking and predictive insights



Response Time

Minutes

Supervisors now react within minutes, not next day



Supervisors now react within minutes, not next day

Technology Stack

A robust, scalable architecture built on industry-leading technologies and best practices for mission-critical manufacturing operations.

IoT Device Integration	IoT Core
Cloud Infrastructure	Google Cloud Platform (GCP)
Processing	Python, SQL
Application Layer	React, Node.js (Full Stack)
Streaming Storage	Real-time DB
Visualization	Integrated UI Panels & Dashboards

